

Siddarth Thota

siddarth.thota.cs@gmail.com | github.com/KingCobra52

EDUCATION

The Pennsylvania State University University Park, PA

Bachelor of Science in Computer Science Expected 2029

Relevant Coursework: Data Structures (CMPSC 132), Introduction to Data Sciences (DS 200), Matrices / Linear Algebra (MATH 220), Calculus II (MATH 141)

In Progress: Calculus and Vector Analysis (MATH 230)

Activities: Association for Computing Machinery (ACM), Penn State Chapter

EXPERIENCE

HackPSU

University Park, PA

Design Team Member

September 2026–Present

- Design high-fidelity Figma systems and user flows for registration, schedules, judging queues, and live event dashboards in collaboration with the Technology team.
- Create venue wayfinding, sponsor graphics, social content, and merchandise for an event planned to serve more than 900 attendees.

TECHNICAL SKILLS

Languages & Web: Python, TypeScript, JavaScript, SQL, HTML, CSS

Frameworks: FastAPI, React Native, Next.js

Data & ML: PostgreSQL, Supabase, Pandas, NumPy, Matplotlib, PyTorch

Testing & Tools: Git, GitHub Actions, pytest, HTTP mocking, unit and integration testing

Backend & Systems: REST APIs, database transactions, TCP sockets, asyncio, multithreading

PROJECTS

Artiste — Music Trading Platform GitHub

- Built an iOS virtual artist trading app nearing App Store release, with JWT authentication, artist discovery, buy/sell orders, portfolio tracking, and transactional safeguards against concurrent overspending and duplicate sales.
- Built idempotent YouTube and Last.fm pipelines for 24 artists, batching up to 50 lookups and database writes per request; scheduled three daily runs with failure isolation, monitoring, and 65 mocked unit tests.

HTTP Server — Built from Raw Sockets GitHub

Python, TCP/IP, ThreadPoolExecutor, asyncio, pytest

- Implemented HTTP request parsing, routing, and response generation directly over TCP sockets; evolved the server from sequential handling to a 32-worker thread pool and an asyncio event loop.
- Added Content-Length headers, connection cleanup, and a 10-second receive timeout; tested routing, response framing, and up to 33 concurrent clients over real sockets.

CampusCabs — Student Transportation Venture

Next.js, React, TypeScript, Supabase

- Designed a campus ride-service pilot for Penn State students, targeting affordable short trips around State College with distance-based fares, manually coordinated drivers, and direct payments to drivers.
- Built a responsive Next.js website to recruit riders and drivers, storing signups in separate Supabase waitlists with form validation, student-specific email fields, and duplicate detection.
- Completed Happy Valley LaunchBox's four-week Idea TestLab at Penn State; used mentor feedback and developed interview guides to refine the pilot around rider affordability, driver earnings, and adoption barriers.